

Stage 1: Time & Frequency Synchronization (PSS/SSS Search)

A
Generate PSS/SSS Sequences

Custom

B
FFT of Rx Samples
GNU Radio Native – FFT

GNU Radio Native

C
PSS Correlation
GNURadioNative–Multiply Conjugate + IFFT or Correlation Estimator

GNU Radio Native

PSS Peak
Detected?

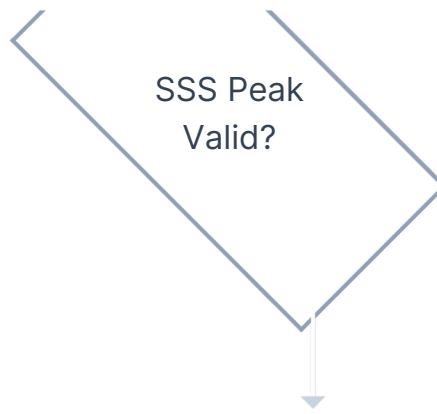
No

Yes

Continue Scanning

D
SSS Correlation
GNURadioNative–Multiply Conjugate + IFFT or Correlation Estimator

GNU Radio Native



No

Yes

Rescan

Custom

Cell ID Resolution: Combine NID2 (from PSS) + NID1 (from SSS) → Full PCI

E

Custom

Compute Frequency Offset & Refinement (PLL/Costas)

Stage 2: PBCH Extraction + Demodulation

This stage focuses on pbch extraction + demodulation. It involves the following key processing blocks:

F

GNU Radio Native

FFT of PBCH Symbols

GNU Radio Native – FFT

G

Custom

PBCH Resource Element Extraction

H

Custom

Channel Estimation from PBCH DMRS

Custom / Native Approximation: SimpleLS can be approximated using Multiply Const, Complex to Float, Complex to Mag blocks; robust MMSE needs custom

I

GNU Radio Native

Equalization

GNURadioNative – Multiply Const, Complex to Float, Complex to Mag

J

GNU Radio Native

QPSK Demodulation → Soft Bits

GNURadioNative–ComplextoFloat+QPSK Demod

Stage 3: PBCH Polar Decoding → MIB

This stage focuses on pbch polar decoding → mib. It involves the following key processing blocks:

K

Custom

PBCH Polar Decoding

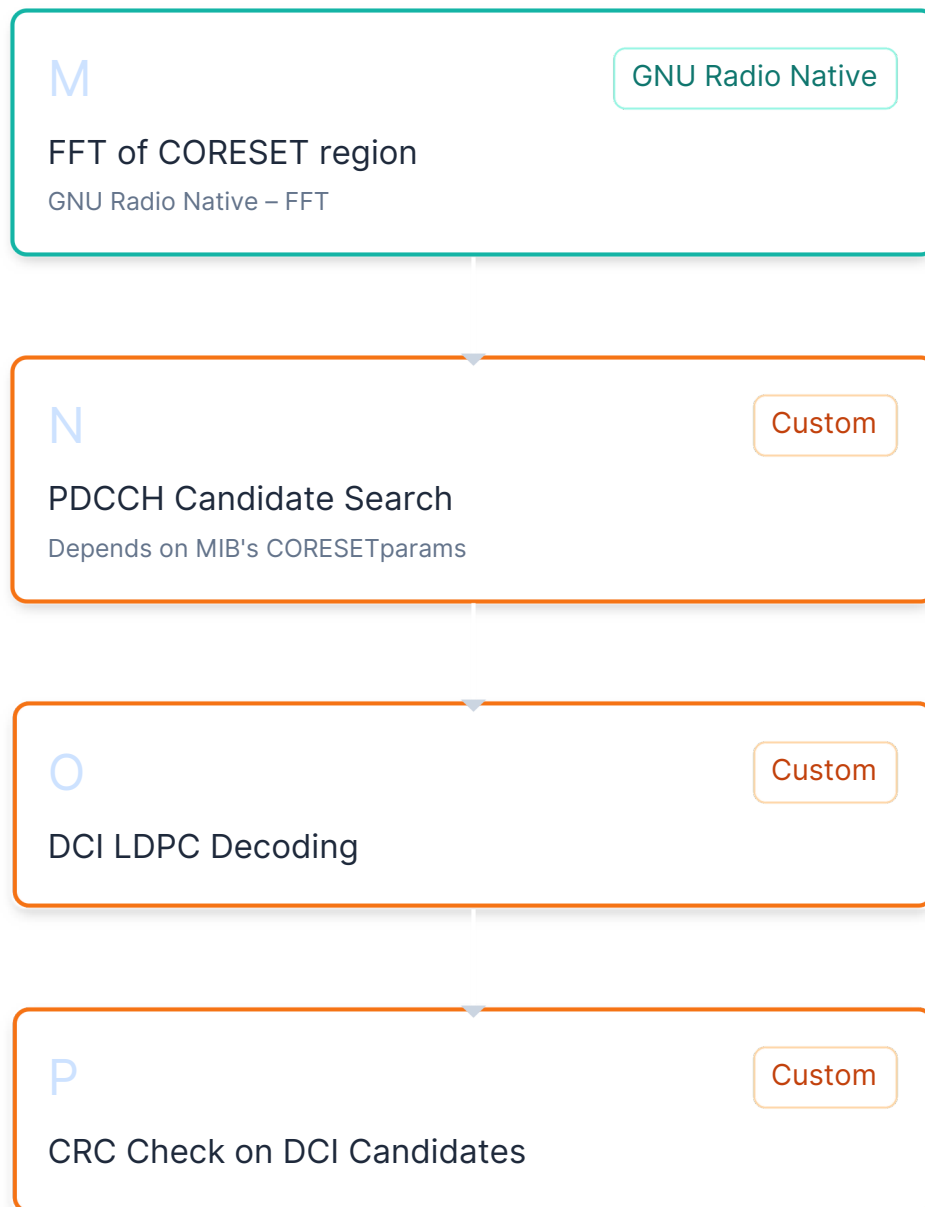
L

Custom

ASN.1 Parsing of MIB

Stage 4: CORESET & PDCCH Decoding

This stage focuses on coreset & pdcch decoding. It involves the following key processing blocks:



Stage 5: SIB1 PDSCH Demodulation + Decoding

This stage focuses on sib1 pdsch demodulation + decoding. It involves the following key processing blocks:

Q

FFT of PDSCH Symbols

GNU Radio Native – FFT

GNU Radio Native

R

PDSCH DMRS Channel Estimation

Custom / NativeApproximation

Custom

S

Equalization

GNURadioNative – Multiply Const, Complex to Float, Complex to Mag

GNU Radio Native

T

QPSK Demodulation → Soft Bits

GNURadioNative–ComplextoFloat+QPSK Demod

GNU Radio Native

U

Rate Matching & Deinterleaving

Custom

V

LDPC Decoding of SIB1

Custom

W

ASN.1 Parsing of SIB1

Custom



Stage 6: System Configuration Application

This stage focuses on system configuration application. It involves the following key processing blocks:

X

Apply System Config & Handover to RRC

Custom